NUMPY

01 IMPORTING NUMPY

```
import numpy as np
```

02 CREATING ARRAYS

From a list:

```
arr = np.array([1, 2, 3, 4, 5])
```

Zeros and ones arrays:

```
zeros_arr = np.zeros(5)
ones_arr = np.ones(5)
```

Range of values:

```
range_arr = np.arange(start, stop, step)
```

Random values:

```
rand_arr = np.random.rand(3, 3)  # 3x3 random
array
```

03 ARRAY OPERATIONS

Basic arithmetic operations:

```
result = arr1 + arr2
result = arr1 - arr2
result = arr1 * arr2
result = arr1 / arr2
```

Element-wise operations:

```
result = np.square(arr)
result = np.sqrt(arr)
result = np.exp(arr)
```

Dot product:

```
dot_product = np.dot(arr1, arr2)
```

04 ARRAY MANIPULATION

Reshape:

```
reshaped_arr = arr.reshape(rows, cols)
```

Transpose:

```
transposed_arr = arr.T
```

Flatten:

```
flattened_arr = arr.flatten()
```

05 STATISTICAL OPERATIONS

Mean, Median, Standard Deviation:

```
mean_val = np.mean(arr)
median_val = np.median(arr)
std_dev = np.std(arr)
```

Sum, Min, Max:

```
total_sum = np.sum(arr) min_val = np.min(arr)
max_val = np.max(arr)
```

06 INDEXING AND SLICING

```
element = arr[index]
sub_array = arr[start:stop]
```

07 LOGICAL OPERATIONS

```
bool_arr = arr > 3
```

08 BROADCASTING

```
result = arr + 5  # Adds 5 to each element of the array
```

09 CONCATENATION

```
combined_arr = np.concatenate((arr1, arr2), axis=0)  #
Concatenate along rows (axis=0)
```

10 STACKING

```
stacked_arr = np.vstack((arr1, arr2))  # Vertically stack
arrays
```

11 LINEAR ALGEBRA

```
eigenvalues, eigenvectors = np.linalg.eig(matrix)
```

12 RANDOM SAMPLING

```
random_sample = np.random.choice(arr, size=3,
replace=False)
```

13 AVOIDING COPY

```
new_arr = arr.copy()
```

14 HANDLING NAN

```
has_nan = np.isnan(arr).any()
```

15 VECTORIZED OPERATIONS

```
result = np.sin(arr)
```

16 SAVE AND LOAD

```
np.save('my_array.npy', arr)
loaded_arr = np.load('my_array.npy')
```

17 MEMORY USAGE

```
size_in_bytes = arr.nbytes
```